

SELECT INVESTIGATION

Clause	Purpose	Syntax	Example	Notes
ORDER BY	Sorts the result rows	ORDER BY column [ASC DESC] [NULLS FIRST NULLS LAST]	SELECT Name, Budget FROM Projects ORDER BY Budget DESC;	ASC is the default. Multiple columns allowed: ORDER BY Department, Name. In PostgreSQL, NULLs go last in ASC and first in DESC by default.
LIMIT	Restricts the number of rows returned	LIMIT n [OFFSET m]	SELECT Name FROM Projects ORDER BY Budget DESC LIMIT 2;	Use with ORDER BY, otherwise the rows returned are unpredictable. OFFSET skips rows (pagination). PostgreSQL also supports the SQL standard FETCH FIRST n ROWS ONLY. SQL Server uses TOP or OFFSET, FETCH instead.
GROUP BY	Groups rows with the same value to apply aggregate functions	GROUP BY column	SELECT DepartmentId, COUNT(*) FROM Employees GROUP BY DepartmentId;	Works with COUNT, SUM, AVG, MIN, MAX. Non-aggregated columns in SELECT must be in GROUP BY, unless grouping by the primary key. Filter groups with HAVING.
INNER JOIN	Returns only rows with a match in both tables	A INNER JOIN B ON A.col = B.col	SELECT e.Name, d.Name FROM Employees e INNER JOIN Departments d ON d.Id = e.DepartmentId;	Equivalent to an intersection. Rows without a match are excluded.
LEFT JOIN	Returns all rows from the left table, plus matches from the right	A LEFT JOIN B ON A.col = B.col	SELECT d.Name, e.Name FROM Departments d LEFT JOIN Employees e ON e.DepartmentId = d.Id;	Right-side columns are NULL when there is no match. Useful to find departments without employees.
RIGHT JOIN	Returns all rows from the right table, plus matches from the left	A RIGHT JOIN B ON A.col = B.col	SELECT d.Name, e.Name FROM Employees e RIGHT JOIN Departments d ON d.Id = e.DepartmentId;	Left-side columns are NULL when there is no match. Any RIGHT JOIN can be rewritten as a LEFT JOIN by swapping the tables, so it is less common in practice.